Book Allocation Problem ⇒

– GfG ⇒ Allocate Minimum Pages

Question Understanding ⇒

$a = [\overset{0}{12}, \overset{1}{34}, \overset{2}{67}, \overset{3}{90}]$ $k = 2$

$a[i]$ ⇒ no of pages in i^{th} book

0th book = 12 pages
— — — so on.

k ⇒ no of students

⇒ Task Allocate books to each student

→ Each student assign at least 1 book
→ " " " contiguous sequence of books

⇒ $[12, 34, 67, 90]$ $\boxed{k = 2}$

output = 113

Permutation of allocations ⇒ $[\underset{12}{12}]$ & $[\underset{191}{34, 67, 90}]$ Max Pages so far 191

→ $[\underset{46}{12, 34}]$ & $[\underset{157}{67, 90}]$ Max Pages so far 157

⇒ $[\underset{113}{12, 34, 67}]$ & $[\underset{90}{90}]$ Max Pages so far 113

⇒ $[12, 34, 67, 90]$ $k = 2$

linear Srch Total = 203

Now, if $[12]$ & $[203 - 12] = [191]$ max = 191

if $[\underset{46}{12 + 34}]$ & $[203 - 46] = [157]$ = 157

if $[\underset{113}{12 + 34 + 67}]$ & $[203 - 113] = [90]$ = 113

$\min[191, 157, 113] = 113$ Ans

Ex $\overset{0}{10} \mid \overset{1}{20} \mid \overset{2}{30} \mid \overset{3}{40}$ $k = 2$

① $10 \mid 20, 30, 40$ = max 90
② $\underset{30}{10, 20} \mid 30, 40$ = max 70
③ $\underset{50}{10, 20, 30} \mid \underset{40}{40}$ = max 60

$\boxed{\text{max} = 60}$

① $0 \text{———} 100$, $M = 50$

I ⇒ $\underset{30}{10 + 20} + \cancel{30} = 60 > M$
II ⇒ $30 + 40 = 70 > M$
III ⇒ $M = 2$

(no possible sol)
start = mid+1

② $51 \text{———} 100$, $\frac{101}{2} = 75$

I ⇒ $\underset{60}{10 + 20 + 30} + 40 = 100 > M$
II ⇒ $40 \Rightarrow 40 < M$ ✓

(possible sol)
{
ans = mid
end = mid-1;
}

③ $51 \text{———} 74$, $\frac{51 + 74}{2} = \frac{125}{2} = 62$

I ⇒ $\underset{60}{10 + 20 + 30} + 40 = 100 > M$
II ⇒ $40 \Rightarrow 40 < M$ ✓

(possible sol)
{
ans = mid
end = mid-1;
}

④ $51 \text{———} 61$, $\frac{51 + 61}{2} = \frac{112}{2} = 56$

I ⇒ $\underset{50}{10 + 20 + 30} + \cancel{40} = \underset{60}{60} > M$
II ⇒ $30 + 40 = 70 > M$
III ⇒ $M = 2$

(no possible sol)
{
start = mid+1;
}

⑤ $57 \text{———} 61$, $\frac{57 + 61}{2} = \frac{118}{2} = 59$

I ⇒ $\underset{50}{10 + 20 + 30} + 40 = 100 > 59$
II ⇒ $\underset{70}{30 + 40} = 70 < M$
III ⇒ $M = 2$

(no possible sol)
{
start = mid+1;
}

⑥ $60 \text{———} 61$ ⇒

I ⇒ $\underset{60}{10 + 20 + 30} + 40 = 100 > M$
II ⇒ $40 \Rightarrow 40 < M$ ✓

(possible sol)
{
ans = mid
end = mid-1
}

⑦ $60 \text{———} 60$

I ⇒ $\underset{60}{10 + 20 + 30} + 40 = 100 > M$
II ⇒ $40 = 40 < M$

(possible sol)
{
ans = mid;
end = mid-1;
}

$S = 60$
$e = 59$ } terminate

now, $\boxed{\text{Ans} = 60}$

⇒ Code ⇒

```
int findPages (vector<int> & arr, int k)
{
    int n = arr.size();
    if ( k > n) return -1, // if student is more than books is less.

    int start = 0
    int end = accumulate(arr.begin(), arr.end(), 0);
    int ans = -1;

    while ( start <= end)
    {
        mid = start + (end - start)/2;

        if ( isPossibleSolution (arr, mid, k))
        {
            ans = mid;
            end = mid-1;
        }
        else { start = mid+1;
        }
    return ans;

bool isPossible() (arr, mid, k)
{
    int Pagesum = 0;
    int Count = 1;

    for( i ⟶ n-1)
    {   if (arr[i] > mid)
        {   return false;
        }
        if ( Pagesum + arr[i] > mid) ⇒ case: 10 | 20 | 30, mid = 50; 10+20 + 30 = 60 > mid; pagesum arr[i]
        {
            count++;
            pagesum = arr[i];
            if ( count > k)
            {   return false;
            }
        }
        else { pagesum += arr[i];
        }
    }
    return true;
```

1 → 2

x 2 > 2

⇒ if initialise count = 0 then (count > k-1)

{ pagesum = 0; pagesum = arr[i] }